

Erdős #396: the antipodal law for shifted windows of $\binom{2n}{n}$

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Abstract

#396 asks whether for every k there is an n with $\prod_{0 \leq i \leq k} (n-i) \mid \binom{2n}{n}$. We give an exact digit-sum criterion for this divisibility, valid for every n and k , and use it to prove an *antipodal law*: for every prime $p > \max(2k, \sqrt{2n})$ the residue $n \bmod p$ decides both the divisor condition and the carry condition, and the two conditions occupy complementary half-turns of the clock $\mathbb{Z}/p$. Consequently a *negative* shift $n-i$ forces $p \nmid \binom{2n}{n}$, while a *positive* shift $n+i$ forces $p \mid \binom{2n}{n}$. This proves that every solution of #396 has all of $n, n-1, \dots, n-k$ free of prime factors above $\max(2k, \sqrt{2n})$, and it explains why Pomerance's companion problem for $\prod_{1 \leq i \leq k} (n+i)$ has density 1 while #396 does not. Two unconditional lower bounds on the least solution follow. #396 itself is not proved.

1 Ledger form

Throughout p is prime, $s_p(m)$ is the sum of the base- p digits of m , and v_p is the p -adic valuation. We use Legendre's formula $v_p(m!) = (m - s_p(m))/(p-1)$ and Kummer's theorem in the form

$$v_p \binom{2n}{n} = \sum_{j \geq 1} \left(\lfloor 2n/p^j \rfloor - 2 \lfloor n/p^j \rfloor \right) = \# \left\{ j \geq 1 : \left\{ n/p^j \right\} \geq \frac{1}{2} \right\},$$

each summand being 0 or 1. We call the indicator $\{n/p^j\} \geq \frac{1}{2}$ the *half-turn at level j* : it says that the point n/p^j lies in the upper half of the clock of circumference 1, equivalently that $n \bmod p^j \geq p^j/2$.

Theorem 1 (exact criterion). *Let $n \geq k+1 \geq 1$. Then*

$$\prod_{0 \leq i \leq k} (n-i) \mid \binom{2n}{n} \iff 3 s_p(n) \geq (k+1) + s_p(n-k-1) + s_p(2n) \text{ for every prime } p.$$

Proof. Since $\prod_{0 \leq i \leq k} (n-i) = n!/(n-k-1)!$, Legendre's formula gives

$$v_p \left(\prod_{0 \leq i \leq k} (n-i) \right) = \frac{(n - s_p(n)) - ((n-k-1) - s_p(n-k-1))}{p-1} = \frac{(k+1) - s_p(n) + s_p(n-k-1)}{p-1},$$

and, again by Legendre,

$$v_p \binom{2n}{n} = \frac{(2n - s_p(2n)) - 2(n - s_p(n))}{p-1} = \frac{2s_p(n) - s_p(2n)}{p-1}.$$

Divisibility holds if and only if the first quantity is at most the second for every p ; clearing the common positive denominator $p - 1$ and rearranging gives the stated inequality. $\square$

Verified: Theorem ?? agrees with direct evaluation of $\binom{2n}{n} \bmod \prod_{0 \leq i \leq k} (n - i)$ for all 1792 pairs with $n < 260$ and $k \leq 6$ — no mismatch.

The criterion is exact and involves no estimate: the whole of the divisibility is the comparison of three digit sums against the window length $k + 1$. Note that at $p = 2$ one has $s_2(2n) = s_2(n)$ and the criterion collapses to a two-term inequality; this is exploited in Corollary ??.

2 The antipodal law

Fix k and n and put

$$y := \max(2k, \sqrt{2n}).$$

For a prime $p > y$ the entire ledger at p is determined by the single residue $n \bmod p$, because $p^2 > 2n$ kills every level $j \geq 2$. The following three statements are then exact descriptions of three arcs of the clock $\mathbb{Z}/p$.

Lemma 2 (the three arcs). *Let $p > y$ and write $\rho = n \bmod p$. Then*

- (a) p divides $\prod_{0 \leq i \leq k} (n - i)$ if and only if $\rho \in [0, k]$, and then $v_p(\prod_{0 \leq i \leq k} (n - i)) = 1$;
- (b) p divides $\prod_{1 \leq i \leq k} (n + i)$ if and only if $\rho \in [p - k, p - 1]$, and then $v_p(\prod_{1 \leq i \leq k} (n + i)) = 1$;
- (c) p divides $\binom{2n}{n}$ if and only if $\rho \geq p/2$, and then $v_p(\binom{2n}{n}) = 1$.

Proof. (a) $p \mid n - i$ for some $0 \leq i \leq k$ means $n \equiv i \pmod{p}$; as $0 \leq i \leq k < p$ this is exactly $\rho \in [0, k]$, and the i is unique since $p > k$. Moreover $p^2 > 2n > n - i$, so $v_p(n - i) = 1$. (b) Likewise $p \mid n + i$ with $1 \leq i \leq k$ means $\rho = p - i$, and $p^2 > 2n \geq n + i$ gives $v_p(n + i) = 1$. (c) By Kummer, $v_p(\binom{2n}{n})$ counts the levels $j \geq 1$ with $\{n/p^j\} \geq \frac{1}{2}$. For $j \geq 2$ we have $p^j \geq p^2 > 2n > n$, so $\{n/p^j\} = n/p^j < \frac{1}{2}$ and the level contributes 0. Only $j = 1$ survives, and $\{n/p\} = \rho/p \geq \frac{1}{2}$ is $\rho \geq p/2$. $\square$

Since $p > 2k$ we have $k < p/2$, so the arc $[0, k]$ of (a) and the arc $[p/2, p)$ of (c) are *disjoint*, while the arc $[p - k, p - 1]$ of (b) is *contained* in the arc of (c). The two shift directions sit at antipodal phases of the same clock. This is the whole content of the next two theorems.

Theorem 3 (negative shifts: extinction). *Let $p > \max(2k, \sqrt{2n})$ be prime. If p divides $n - i$ for some $0 \leq i \leq k$, then $v_p(\binom{2n}{n}) = 0$. Consequently, if $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ then no prime exceeding $\max(2k, \sqrt{2n})$ divides any of $n, n - 1, \dots, n - k$.*

Proof. By Lemma ??(a) the hypothesis gives $\rho = n \bmod p \in [0, k]$, and $k < p/2$ because $p > 2k$; so $\rho < p/2$ and Lemma ??(c) gives $v_p(\binom{2n}{n}) = 0$. If the divisibility held we would need $1 = v_p(n - i) \leq v_p(\binom{2n}{n}) = 0$, a contradiction. $\square$

Corollary 4 (the smooth-window requirement). *If $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ then every one of the $k + 1$ consecutive integers $n - k, n - k + 1, \dots, n$ is y -smooth, where $y = \max(2k, \sqrt{2n})$.*

Proof. Immediate from Theorem ?? : a prime factor of any $n - i$ exceeding y is excluded. $\square$

Verified: over all solutions (n, k) with $n < 400$ and $k \leq 7$, all 111 prime factors of the $k + 1$ window entries are $\leq y$ — no exception. Separately, over $n < 1500$ and $k \leq 8$, all 42 112 prime factors exceeding y of window entries $n - i$ occur at non-solutions and every one of them satisfies $v_p\binom{2n}{n} = 0$, as Theorem ?? requires.

Theorem 5 (positive shifts: automatic). *Let $p > \max(2k, \sqrt{2n})$ be prime and let $1 \leq i \leq k$. If p divides $n + i$ then $v_p(n + i) = 1 \leq v_p\binom{2n}{n}$. Consequently the divisibility $\prod_{1 \leq i \leq k} (n + i) \mid \binom{2n}{n}$ imposes no condition whatsoever at primes above $\max(2k, \sqrt{2n})$: it is decided entirely by the primes $p \leq \max(2k, \sqrt{2n})$, and in particular it places no smoothness requirement on the window.*

Proof. By Lemma ??(b), $\rho = n \bmod p = p - i$ and $v_p(n + i) = 1$. Since $i \leq k < p/2$ we get $\rho = p - i > p/2$, so Lemma ??(c) gives $v_p\binom{2n}{n} = 1$. $\square$

Verified: over $n < 1500$ and $k \leq 8$ there are 33 752 pairs $(n + i, p)$ with $p > y$ and $p \mid n + i$; in every one $v_p(n + i) = 1$ and $v_p\binom{2n}{n} = 1$ — no exception.

Theorems ?? and ?? are the same computation read at two antipodal phases, and together they account for the asymmetry in the literature. Pomerance showed that the set of n with $\prod_{1 \leq i \leq k} (n + i) \mid \binom{2n}{n}$ has density 1; by Theorem ?? that statement is a statement about the finitely many primes below $\max(2k, \sqrt{2n})$ only, where the digit-sum criterion of Theorem ?? is satisfied for almost all n . By contrast Corollary ?? shows that #396 cannot be attacked prime by prime at all: it demands a block of $k + 1$ consecutive integers with no prime factor above $\sqrt{2n}$, an event of density $\asymp \rho(2)^{k+1}$ in the Dickman function, and this — not any local carry condition — is what makes #396 hard.

3 Two unconditional lower bounds on a solution

Corollary 6 (binary bound). *If $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ then*

$$2s_2(n) \geq (k + 1) + s_2(n - k - 1), \quad \text{hence} \quad s_2(n) \geq \left\lceil \frac{k+1}{2} \right\rceil \quad \text{and} \quad n \geq 2^{(k-1)/2}.$$

Proof. Apply Theorem ?? at $p = 2$ and use $s_2(2n) = s_2(n)$ to cancel one copy of $s_2(n)$. The second inequality drops the nonnegative term $s_2(n - k - 1)$, and the third follows from $s_2(n) \leq \lfloor \log_2 n \rfloor + 1$. $\square$

Verified: all 329 solutions with $n < 3000$, $k \leq 7$ satisfy the two-term inequality — no exception.

Corollary 7 (prime-count bound). *If $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ and $y = \max(2k, \sqrt{2n})$ then*

$$\pi(y) \geq (k + 1) \frac{\log(n - k)}{\log(2n)}.$$

In particular $\pi(y) \geq k$ once $n \geq (2n)^{k/(k+1)} + k$, so y is at least the k -th prime; when the maximum defining y is attained by $\sqrt{2n}$ this gives $n \geq \frac{1}{2}p_k^2$.

Proof. By Corollary ?? the product $\prod_{0 \leq i \leq k} (n - i)$ is y -smooth, and it divides $\binom{2n}{n}$, so it divides $\prod_{p \leq y} p^{v_p(\binom{2n}{n})}$. Kummer's theorem bounds each carry count by $v_p(\binom{2n}{n}) \leq \lfloor \log_p(2n) \rfloor$, i.e. $p^{v_p(\binom{2n}{n})} \leq 2n$. Hence

$$(n - k)^{k+1} < \prod_{0 \leq i \leq k} (n - i) \leq (2n)^{\pi(y)},$$

and taking logarithms gives the claim. $\square$

Verified: at the least solution for $k = 4$, namely $n = 45153$, one has $y = 300$, $\pi(300) = 62$ and the right-hand side is 4.70.

4 Census

Let n_k denote the least n with $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ (OEIS A375077). Searching by the criterion of Theorem ??, with the window prefiltered by Corollary ??, gives

$$n_0 = 2, \quad n_1 = 210, \quad n_2 = 2480, \quad n_3 = 8178, \quad n_4 = 45153, \quad n_5 > 3 \times 10^6.$$

Every one of these has its whole window y -smooth, as Corollary ?? forces: for instance at $k = 1$, $n_1 = 210 = 2 \cdot 3 \cdot 5 \cdot 7$ and $209 = 11 \cdot 19$, with $y = \max(2, \lfloor \sqrt{420} \rfloor) = 20$, and indeed $19 \leq 20$. The growth of n_k is far faster than the bounds of Corollaries ?? and ?? ($2^{(k-1)/2}$ is 2.8 at $k = 4$, against 45153), which is consistent with Corollary ??: the binding constraint is the smooth window, not any local carry budget.

Remark 8 (harmonic content). *The object's own scale is the clock $\mathbb{Z}/p^j$, never unit 1; the DC mode is the Legendre digit-sum identity, which is evaluated exactly in Theorem ?? with no remainder to estimate, and the entire AC content at a prime $p > y$ is the single half-turn indicator $\mathbf{1}[\rho \geq p/2]$. The difficulty of #396 lives on the carrier — the requirement that a whole window of length $k + 1$ be $\sqrt{2n}$ -smooth — and not in any chart readout.*

Remark 9 (register). **Proved:** Theorem ?? (exact digit-sum criterion, all n, k); Lemma ??; Theorem ?? and Corollary ?? (every solution of #396 has a $\max(2k, \sqrt{2n})$ -smooth window); Theorem ?? (the positive-shift problem has no large-prime condition at all); Corollaries ?? and ??. All are elementary consequences of Legendre and Kummer, and all were checked against direct computation before being written down. **Measured:** $n_0, \dots, n_4$ and $n_5 > 3 \times 10^6$; the verification counts recorded after each statement. **Not proved:** #396 itself. What Corollary ?? leaves is the existence, for every k , of $k + 1$ consecutive $\sqrt{2n}$ -smooth integers additionally satisfying the digit-sum criterion at every prime below $\sqrt{2n}$; no construction of such blocks is given here, and none is claimed. **Falsifier:** a solution (n, k) with a prime factor of some $n - i$ exceeding $\max(2k, \sqrt{2n})$ would refute Theorem ??; none was found in 42 112 tested large-prime incidences.